Лабораторная работа

Номер 8

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Информация

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Цель работы

Получить практические навыки установки и настройки SMTP-сервера

Установка Postfix

Сначала заходим на сервер под root и ставим postfix

```
[sudo] password for dvkalashnikova:
[root@server.dvkalashnikova.net ~]# dnf -y install postfix
Extra Packages for Enterprise Linux 10 - x86_64                  24 kB/s |  38 kB     00:01
Extra Packages for Enterprise Linux 10 - x86_64                 688 kB/s | 6.7 MB     00:09
Rocky Linux 10 - BaseOS                                         5.2 kB/s | 4.3 kB     00:00
Rocky Linux 10 - BaseOS                                         2.6 MB/s |  38 MB     00:14
Rocky Linux 10 - AppStream                                      5.0 kB/s | 4.4 kB     00:00
Rocky Linux 10 - AppStream                                      262 kB/s | 2.8 MB     00:10
Rocky Linux 10 - CRB                                            2.9 kB/s | 4.3 kB     00:01
Rocky Linux 10 - CRB                                            166 kB/s | 559 kB     00:03
Rocky Linux 10 - Extras                                         1.6 kB/s | 3.1 kB     00:01
Rocky Linux 10 - Extras                                         4.9 kB/s | 6.1 kB     00:01
Last metadata expiration check: 0:00:02 ago on Mon 19 Oct 2026 07:48:43 PM UTC.
Dependencies resolved.
================================================================================================================
 Package                 Architecture          Version                  Repository              Size
================================================================================================================
Installing:
 postfix                 x86_64                2:3.8.5-10.el10_2        appstream              1.5 M
Installing dependencies:
 postfix-lmdb            x86_64                2:3.8.5-10.el10_2        appstream               23 k

Transaction Summary
================================================================================================================
Install  2 Packages

Total download size: 1.5 M
Installed size: 4.5 M
Downloading Packages:
(1/2): postfix-lmdb-3.8.5-10.el10_2.x86_64.rpm                   38 kB/s |  23 kB     00:00
(2/2): postfix-3.8.5-10.el10_2.x86_64.rpm                       1.1 MB/s | 1.5 MB     00:01
----------------------------------------------------------------------------------------------------------------
Total                                                           888 kB/s | 1.5 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                        1/1
  Installing       : postfix-lmdb-2:3.8.5-10.el10_2.x86_64                                  1/2
  Running scriptlet: postfix-2:3.8.5-10.el10_2.x86_64                                       2/2
  Installing       : postfix-2:3.8.5-10.el10_2.x86_64                                       2/2
  Running scriptlet: postfix-2:3.8.5-10.el10_2.x86_64                                       2/2

Installed:
  postfix-2:3.8.5-10.el10_2.x86_64                 postfix-lmdb-2:3.8.5-10.el10_2.x86_64

Complete!
[root@server.dvkalashnikova.net ~]#
```

Рис. 1: Установка Postfix

Установка s-nail

Следом устанавливаем утилиту s-nail

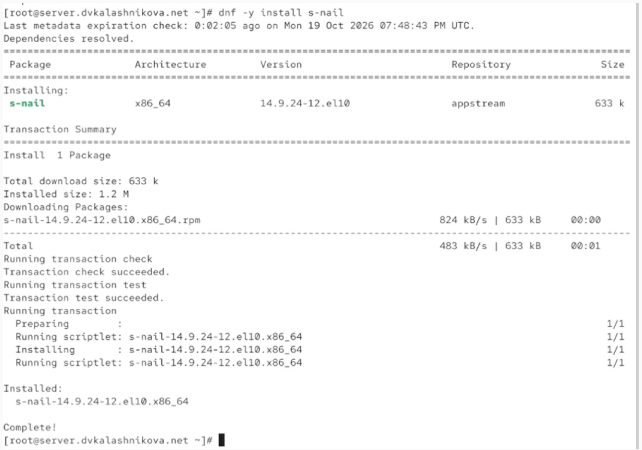


Рис. 2: Установка s-nail

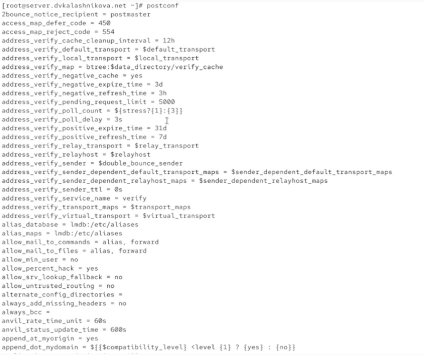
Предварительная настройка

Открываем в фаерволе доступ к smtp, восстанавливаем контексты selinux и запускаем postfix

```
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=smtp
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https smtp ssh ssh-custom
[root@server.dvkalashnikova.net ~]# restorecon -vR /etc
[root@server.dvkalashnikova.net ~]# systemctl enable postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/
postfix.service'.
[root@server.dvkalashnikova.net ~]# systemctl start postfix
[root@server.dvkalashnikova.net ~]#
```

Рис. 3: Предварительная настройка

Выполняем postconf — команда выводит полный список параметров конфигурации



```
[root@server.dvkalashnikova.net ~]# postconf
2bounce_notice_recipient = postmaster
access_map_defer_code = 450
access_map_reject_code = 554
address_verify_cache_cleanup_interval = 12h
address_verify_default_transport = $default_transport
address_verify_local_transport = $local_transport
address_verify_map = btree:$data_directory/verify_cache
address_verify_negative_cache = yes
address_verify_negative_expire_time = 3d
address_verify_negative_refresh_time = 3h
address_verify_pending_request_limit = 5000
address_verify_poll_count = ${stress?{1}:{3}}
address_verify_poll_delay = 3s
address_verify_positive_expire_time = 31d
address_verify_positive_refresh_time = 7d
address_verify_relay_transport = $relay_transport
address_verify_relayhost = $relayhost
address_verify_sender = $double_bounce_sender
address_verify_sender_dependent_default_transport_maps = $sender_dependent_default_transport_maps
address_verify_sender_dependent_relayhost_maps = $sender_dependent_relayhost_maps
address_verify_sender_ttl = 0s
address_verify_service_name = verify
address_verify_transport_maps = $transport_maps
address_verify_virtual_transport = $virtual_transport
alias_database = lmdb:/etc/aliases
alias_maps = lmdb:/etc/aliases
allow_mail_to_commands = alias, forward
allow_mail_to_files = alias, forward
allow_min_user = no
allow_percent_hack = yes
allow_srv_lookup_fallback = no
allow_untrusted_routing = no
alternate_config_directories =
always_add_missing_headers = no
always_bcc =
anvil_rate_time_unit = 60s
anvil_status_update_time = 600s
append_at_myorigin = yes
append_dot_mydomain = ${{$compatibility_level} <level {1} ? {yes} : {no}}
```

Рис. 4: postconf

Обновление конфигурации postfix

Смотрим значения myorigin и mydomain. Меняем myorigin на доменное имя, проверяем через postfix check и перезапускаем службу

```
[root@server.dvkalashnikova.net ~]# postconf myorigin
myorigin = $myhostname
[root@server.dvkalashnikova.net ~]# postconf mydomain
mydomain = dvkalashnikova.net
[root@server.dvkalashnikova.net ~]# postconf -e 'myorigin = $mydoma''''''''''''''
''''''
postconf: fatal: missing '=' after attribute name: "???myorigin"
[root@server.dvkalashnikova.net ~]# postconf -e 'myorigin = $mydomain'
>
> ^C
[root@server.dvkalashnikova.net ~]# postconf -e 'myorigin = dvkalashnikova.net'
> ^C
[root@server.dvkalashnikova.net ~]# postconf -e 'myorigin = $mydomain'
[root@server.dvkalashnikova.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.dvkalashnikova.net ~]# postfix check
[root@server.dvkalashnikova.net ~]# systemctl reload postfix
```

Рис. 5: Обновление конфигурации postfix

postconf -n

Командой postconf -n выводим только те настройки, которые отличаются от значений по умолчанию

```
[root@server.dvkalashnikova.net ~]# postconf -n
alias_database = lmdb:/etc/aliases
alias_maps = lmdb:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 3.8
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name $
process_id & sleep 5
default_database_type = lmdb
html_directory = no
inet_interfaces = localhost
inet_protocols = all
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
myorigin = $mydomain
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
smtp_tls_CApath = /etc/pki/tls/certs
smtp_tls_security_level = may
smtpd_tls_cert_file = /etc/pki/tls/certs/postfix.pem
smtpd_tls_key_file = /etc/pki/tls/private/postfix.key
smtpd_tls_security_level = may
unknown_local_recipient_reject_code = 550
[root@server.dvkalashnikova.net ~]#
```

Рис. 6: postconf -n

Настройка протоколов

Жёстко фиксируем имя домена. Оставляем только ipv4 и перезапускаем postfix

```
[root@server.dvkalashnikova.net ~]# postconf -e 'mydomain = dvkalashnikova.net'
[root@server.dvkalashnikova.net ~]# postconf inet_protocols
inet_protocols = all
[root@server.dvkalashnikova.net ~]# postconf -e 'inet_protocols = ipv4'
[root@server.dvkalashnikova.net ~]# postfix check
[root@server.dvkalashnikova.net ~]# systemctl reload postfix
```

Рис. 7: Обновление настроек протоколов

Отправка письма

Пробуем отправить письмо самому себе со своей учётной записи

```
[root@server.dvkalashnikova.net ~]# echo .| mail -s test1 dvkalashnikova@server.dvkalashnikova.net
[root@server.dvkalashnikova.net ~]#
```

Рис. 8: Отправка письма

Лог почты

В логе из /var/log/maillog видим «delivered to mailbox» — письмо доставлено

```
[dvkalashnikova@server.dvkalashnikova.net ~]$ sudo -i
[sudo] password for dvkalashnikova:
[root@server.dvkalashnikova.net ~]# tail -f /var/log/maillog
Oct 19 20:15:44 server postfix/postfix-script[16647]: refreshing the Postfix mail system
Oct 19 20:15:44 server postfix/master[13789]: reload -- version 3.8.5, configuration /etc/
postfix
Oct 19 20:15:44 server postfix/master[13789]: warning: ignoring inet_protocols parameter v
alue change
Oct 19 20:15:44 server postfix/master[13789]: warning: old value: "all", new value: "ipv4"
Oct 19 20:15:44 server postfix/master[13789]: warning: to change inet_protocols, stop and
start Postfix
Oct 19 20:16:45 server postfix/pickup[16652]: 2F895412619B: uid=0 from=<root>
Oct 19 20:16:45 server postfix/cleanup[16797]: 2F895412619B: message-id=<20261019201645.2F
895412619B@server.dvkalashnikova.net>
Oct 19 20:16:45 server postfix/qmgr[16651]: 2F895412619B: from=<root@dvkalashnikova.net>,
size=370, nrcpt=1 (queue active)
Oct 19 20:16:45 server postfix/local[16799]: 2F895412619B: to=<dvkalashnikova@server.dvkal
ashnikova.net>, relay=local, delay=0.18, delays=0.1/0.06/0/0.01, dsn=2.0.0, status=sent (d
elivered to mailbox)
Oct 19 20:16:45 server postfix/qmgr[16651]: 2F895412619B: removed
```

Рис. 9: Лог почты

/var/spool/mail

Проверяем, что в /var/spool/mail появились файлы пользователей

```
[root@server.dvkalashnikova.net ~]# sudo ls /var/spool/mail/
dvkalashnikova  vagrant
```

Рис. 10: /var/spool/mail

Установка на клиенте

Переходим на клиентскую машину, заходим под root и тоже ставим postfix

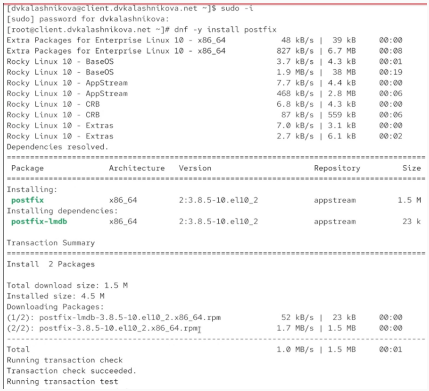


Рис. 11: Установка Postfix

Установка s-nail на клиенте

Ставим и s-nail

```
[root@client.dvkalashnikova.net ~]# dnf -y install s-nail
Waiting for process with pid 14538 to finish.
Last metadata expiration check: 0:00:02 ago on Mon 19 Oct 2026 08:28:08 PM UTC.
Dependencies resolved.
================================================================================
 Package          Architecture     Version                    Repository      Size
================================================================================
Installing:
 s-nail           x86_64           14.9.24-12.el10            appstream      633 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 633 k
Installed size: 1.2 M
Downloading Packages:
s-nail-14.9.24-12.el10.x86_64.rpm                 1.1 MB/s | 633 kB     00:00
--------------------------------------------------------------------------------
Total                                             521 kB/s | 633 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64                          1/1
  Installing       : s-nail-14.9.24-12.el10.x86_64                          1/1
  Running scriptlet: s-nail-14.9.24-12.el10.x86_64                          1/1

Installed:
  s-nail-14.9.24-12.el10.x86_64

Complete!
```

Рис. 12: Установка s-nail

Настройка протоколов на клиенте

Повторяем действия с сервера: оставляем только ipv4 и перезапускаем postfix

```
[root@client.dvkalashnikova.net ~]# postconf inet_protocols
inet_protocols = ipv4
[root@client.dvkalashnikova.net ~]# postconf -e 'inet_protocols = ipv4'
[root@client.dvkalashnikova.net ~]# systemctl enable postfix
[root@client.dvkalashnikova.net ~]# systemctl start postfix
[root@client.dvkalashnikova.net ~]#
```

Рис. 13: Настройка протоколов на клиенте

Отправка с клиента

Отправляем с клиента письмо на сервер

```
[dvkalashnikova@client.dvkalashnikova.net ~]$ echo .| mail -s test1 dvkalashnikova@server.dvkalashnikova.
net
[dvkalashnikova@client.dvkalashnikova.net ~]$
```

Рис. 14: Отправка письма

Логи на сервере

На сервере в логах ничего нового не появляется



Рис. 15: Логи почты

Разрешение писем из локальной сети

Меняем inet_interfaces с localhost на all, добавляем в mynetworks 192.168.0.0 и перезапускаем postfix

```
[root@server.dvkalashnikova.net ~]# postconf inet_interfaces
inet_interfaces = localhost
[root@server.dvkalashnikova.net ~]# postconf mynetworks
mynetworks = 127.0.0.1/32
[root@server.dvkalashnikova.net ~]# postconf -e 'inet_interfaces = all'
[root@server.dvkalashnikova.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.dvkalashnikova.net ~]# postfix check
[root@server.dvkalashnikova.net ~]# systemctl reload postfix
[root@server.dvkalashnikova.net ~]# systemctl stop postfix
[root@server.dvkalashnikova.net ~]# systemctl start postfix
```

Рис. 16: Включение получения писем из локальной сети

Повторная отправка

Снова пробуем отправить письмо с клиента

```
[dvkalashnikova@client.dvkalashnikova.net ~]$ echo .| mail -s test1 dvkalashnikova@server.dvkalashnikova.
net
You have new mail in /var/spool/mail/dvkalashnikova
[dvkalashnikova@client.dvkalashnikova.net ~]$
```

Рис. 17: Повторная отправка письма

Успешная доставка

В логах новое письмо EBE862044889, в предпоследней строке — delivered to mailbox

```
[dvkalashnikova@client.dvkalashnikova.net ~]$ sudo -i
[sudo] password for dvkalashnikova:
[root@client.dvkalashnikova.net ~]# tail -f /var/log/maillog
Oct 19 20:50:51 client postfix/pickup[1880]: ADE2540A017C: uid=1001 from=<dvkalashnikova>
Oct 19 20:50:51 client postfix/cleanup[11481]: ADE2540A017C: message-id=<20261019205051.ADE2540A017C@clie
nt.dvkalashnikova.net>
Oct 19 20:50:51 client postfix/qmgr[1881]: ADE2540A017C: from=<dvkalashnikova@client.dvkalashnikova.net>,
 size=370, nrcpt=1 (queue active)
Oct 19 20:50:51 client postfix/smtp[11485]: ADE2540A017C: to=<dvkalashnikova@dvkalashnikova.net>, relay=n
one, delay=0.3, delays=0.14/0.08/0.08/0, dsn=5.4.4, status=bounced (Host or domain name not found. Name s
ervice error for name=dvkalashnikova.net type=A: Host not found)
Oct 19 20:50:51 client postfix/cleanup[11481]: EDBA640A017F: message-id=<20261019205051.EDBA640A017F@clie
nt.dvkalashnikova.net>
Oct 19 20:50:51 client postfix/bounce[11486]: ADE2540A017C: sender non-delivery notification: EDBA640A017
F
Oct 19 20:50:51 client postfix/qmgr[1881]: EDBA640A017F: from=<>, size=2615, nrcpt=1 (queue active)
Oct 19 20:50:51 client postfix/qmgr[1881]: ADE2540A017C: removed
Oct 19 20:50:52 client postfix/local[11487]: EDBA640A017F: to=<dvkalashnikova@client.dvkalashnikova.net>,
 relay=local, delay=0.06, delays=0.03/0.02/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 19 20:50:52 client postfix/qmgr[1881]: EDBA640A017F: removed
```

Рис. 18: Успешная отправка письма согласно логам

Письмо пользователю

Пробуем отправить письмо не на устройство, а пользователю



Рис. 19: Отправка письма пользователю

Неуспешная доставка

Письмо 7482D204488 не доставляется — No route to host

```
[root@server.dvkalashnikova.net ~]# tail -f /var/log/maillog
Oct 19 20:40:46 server postfix/master[13789]: reload -- version 3.8.5, configuration /etc/
postfix
Oct 19 20:40:46 server postfix/master[13789]: warning: ignoring inet_protocols parameter v
alue change
Oct 19 20:40:46 server postfix/master[13789]: warning: old value: "all", new value: "ipv4"
Oct 19 20:40:46 server postfix/master[13789]: warning: to change inet_protocols, stop and
start Postfix
Oct 19 20:40:46 server postfix/master[13789]: warning: service smtp: ignoring inet_interfa
ces change
Oct 19 20:40:46 server postfix/master[13789]: warning: to change inet_interfaces, stop and
 start Postfix
Oct 19 20:40:53 server postfix/postfix-script[20201]: stopping the Postfix mail system
Oct 19 20:40:53 server postfix/master[13789]: terminating on signal 15
Oct 19 20:41:01 server postfix/postfix-script[20310]: starting the Postfix mail system
Oct 19 20:41:01 server postfix/master[20312]: daemon started -- version 3.8.5, configurati
on /etc/postfix
```

Рис. 20: Неуспешная доставка письма

postqueue -p

Информацию о зависшем письме смотрим командой postqueue -p



Рис. 21: postqueue -p

Файл прямой зоны

Правим файл прямой зоны — добавляем запись о почтовом сервере

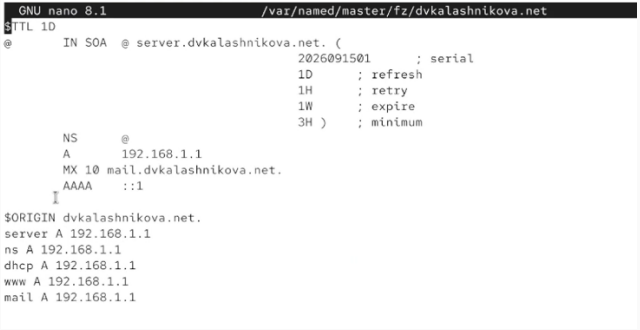


Рис. 22: Файл прямой зоны

Файл обратной зоны

Соответствующие изменения в обратной зоне

```
 GNU nano 8.1                          /var/named/master/rz/192.168.1
$TTL 1D
@       IN SOA  @ server.dvkalashnikova.net. (
                                        2026091501      ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      @
        A       192.168.1.1
        AAAA    ::1
        MX 10 mail.dvkalashnikova.net.
        PTR     server.dvkalashnikova.net
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.dvkalashnikova.net.
1 PTR ns.dvkalashnikova.net.
1 PTR dhcp.dvkalashnikova.net.
1 PTR www.dvkalashnikova.net.
1 PTR mail.dvkalashnikova.net.
```

Рис. 23: Файл обратной зоны

Обновление списков назначения

Обновляем список, перезапускаем postfix, selinux и dns, затем postqueue -f

```
[root@server.dvkalashnikova.net ~]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localh
ost, $mydomain'
[root@server.dvkalashnikova.net ~]# postfix check
[root@server.dvkalashnikova.net ~]# systemctl reload postfix
[root@server.dvkalashnikova.net ~]# restorecon -vR /etc
[root@server.dvkalashnikova.net ~]# restorecon -vR /var/named
[root@server.dvkalashnikova.net ~]# systemctl restart named
[root@server.dvkalashnikova.net ~]# postqueue -f
[root@server.dvkalashnikova.net ~]#
```

Рис. 24: Обновление списков назначения почты

Письмо доставлено

В логах видно, что письмо доставлено

```
[root@server.dvkalashnikova.net ~]# sudo tail -f /var/log/maillog
Oct 19 20:40:46 server postfix/master[13789]: reload -- version 3.8.5, configuration /etc/
postfix
Oct 19 20:40:46 server postfix/master[13789]: warning: ignoring inet_protocols parameter v
alue change
Oct 19 20:40:46 server postfix/master[13789]: warning: old value: "all", new value: "ipv4"
Oct 19 20:40:46 server postfix/master[13789]: warning: to change inet_protocols, stop and
start Postfix
Oct 19 20:40:46 server postfix/master[13789]: warning: service smtp: ignoring inet_interfa
ces change
Oct 19 20:40:46 server postfix/master[13789]: warning: to change inet_interfaces, stop and
 start Postfix
Oct 19 20:40:53 server postfix/postfix-script[20201]: stopping the Postfix mail system
Oct 19 20:40:53 server postfix/master[13789]: terminating on signal 15
Oct 19 20:41:01 server postfix/postfix-script[20310]: starting the Postfix mail system
Oct 19 20:41:01 server postfix/master[20312]: daemon started -- version 3.8.5, configurati
on /etc/postfix
Oct 19 21:01:13 server postfix/postfix-script[23511]: refreshing the Postfix mail system
Oct 19 21:01:13 server postfix/master[20312]: reload -- version 3.8.5, configuration /etc/
postfix
```

Рис. 25: Лог об успешно доставленном письме

Сохранение конфигурации

Сохраняем конфигурацию для vagrant с сервера и создаём mail.sh

```
[root@server.dvkalashnikova.net ~]# cd /vagrant/provision/server/dns/var/named
[root@server.dvkalashnikova.net named]# cp -R /var/named/* /vagrant/provision/server/dns/var/named
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/dvkalashnikova.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol error
[root@server.dvkalashnikova.net named]# cp -R /var/named/* /vagrant/provision/server/dns/var/named
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run-20261005'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run-20261019'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/dvkalashnikova.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol error
[root@server.dvkalashnikova.net named]# cd /vagrant/provision/server
[root@server.dvkalashnikova.net server]# touch mail.sh
[root@server.dvkalashnikova.net server]# chmod +x mail.sh
[root@server.dvkalashnikova.net server]# nano mail.sh
[root@server.dvkalashnikova.net server]#
```

Рис. 26: Сохранение конфигурации

В mail.sh записываем скрипт

```
 GNU nano 8.1                                    mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
systemctl stop postfix
systemctl start postfix
```

Рис. 27: mail.sh

mail.sh для клиента

Переключаемся на клиент и тоже создаём mail.sh

```
[root@client.dvkalashnikova.net ~]# cd /vagrant/provision/client
[root@client.dvkalashnikova.net client]# touch mail.sh
[root@client.dvkalashnikova.net client]# chmod +x mail.sh
[root@client.dvkalashnikova.net client]# nano mail.sh
```

Рис. 28: Создание скрипта mail.sh для клиента

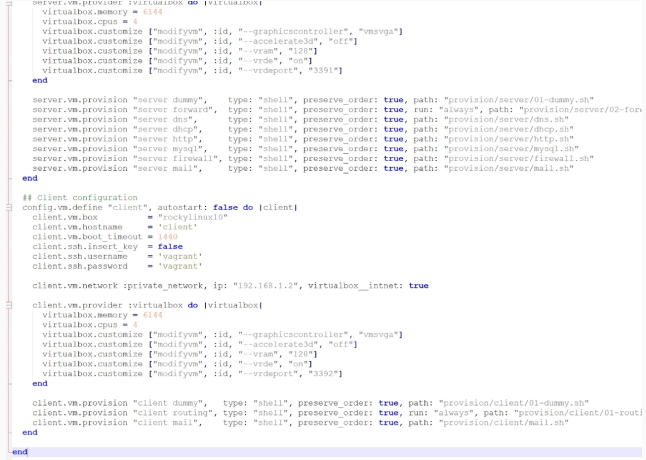
Скрипт для клиента

Скрипт для клиента выглядит так

```
 GNU nano 8.1                                    mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
```

Рис. 29: mail.sh для клиента

В vagrantfile добавляем запуск скриптов для клиента и сервера



```
  server.vm.provider :virtualbox do |virtualbox|
    virtualbox.memory = 6144
    virtualbox.cpus = 4
    virtualbox.customize ["modifyvm", :id, "--graphicscontroller", "vmsvga"]
    virtualbox.customize ["modifyvm", :id, "--accelerate3d", "off"]
    virtualbox.customize ["modifyvm", :id, "--vram", "128"]
    virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
    virtualbox.customize ["modifyvm", :id, "--vrdeport", "3391"]
  end

  server.vm.provision "server dummy",    type: "shell", preserve_order: true, path: "provision/server/01-dummy.sh"
  server.vm.provision "server forward",  type: "shell", preserve_order: true, run: "always", path: "provision/server/02-forw
  server.vm.provision "server dns",      type: "shell", preserve_order: true, path: "provision/server/dns.sh"
  server.vm.provision "server dhcp",     type: "shell", preserve_order: true, path: "provision/server/dhcp.sh"
  server.vm.provision "server http",     type: "shell", preserve_order: true, path: "provision/server/http.sh"
  server.vm.provision "server mysql",    type: "shell", preserve_order: true, path: "provision/server/mysql.sh"
  server.vm.provision "server firewall", type: "shell", preserve_order: true, path: "provision/server/firewall.sh"
  server.vm.provision "server mail",     type: "shell", preserve_order: true, path: "provision/server/mail.sh"
end

## Client configuration
config.vm.define "client", autostart: false do |client|
  client.vm.box              = "rockylinux10"
  client.vm.hostname         = 'client'
  client.vm.boot_timeout = 1440
  client.ssh.insert_key  = false
  client.ssh.username        = 'vagrant'
  client.ssh.password        = 'vagrant'

  client.vm.network :private_network, ip: "192.168.1.2", virtualbox__intnet: true

  client.vm.provider :virtualbox do |virtualbox|
    virtualbox.memory = 6144
    virtualbox.cpus = 4
    virtualbox.customize ["modifyvm", :id, "--graphicscontroller", "vmsvga"]
    virtualbox.customize ["modifyvm", :id, "--accelerate3d", "off"]
    virtualbox.customize ["modifyvm", :id, "--vram", "128"]
    virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
    virtualbox.customize ["modifyvm", :id, "--vrdeport", "3392"]
  end

  client.vm.provision "client dummy",   type: "shell", preserve_order: true, path: "provision/client/01-dummy.sh"
  client.vm.provision "client routing", type: "shell", preserve_order: true, run: "always", path: "provision/client/01-routi
  client.vm.provision "client mail",    type: "shell", preserve_order: true, path: "provision/client/mail.sh"
end
end
```

Рис. 30: vagrantfile

Выводы

В ходе лабораторной работы получены навыки настройки SMTP-сервера

Список литературы

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